



PIXLOOP线控底盘HOOKE

PIXLOOP DRIVE-BY-WIRE Chassis HOOKE

用户使用说明书/User manual



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I. 前言Forword

非常感谢您选择我们 PIX 的产品，您的认可和满意是我们最大的动力。

Thank you very much for choosing our PIX products, your recognition and satisfaction are our greatest motivation.

PIXLOOP-HOOKE 线控底盘是PIX Moving 为工程师打造的一款纯线控底盘，针对园区内 L4级低速自动驾驶提供的线控底盘。该底盘采用了全新的转向模式和新的驱动方式。整车电力由一块72V的动力电池和一个12V免维护铅酸电池共同提供。全车采用高强度钢材搭建。通过遥控器可以轻松控制底盘的转向与动力，同时 PIXLOOP 底盘可自行搭载自动驾驶系统。本文将以HOOKE4轮4驱版本详细介绍底盘每一个功能的同时让你能快速使用本产品。

PIXLOOP HOOKE drive-by-wire chassis is a pure drive-by-wire chassis created by PIX Moving for engineers. It is a drive-by-wire chassis for L4 low-speed autonomous driving in the parks. The chassis features a new steering pattern and a new way of driving. The power of the whole chassis is provided by a 72V power battery and a 12V maintenance-free lead-acid battery. The whole chassis is constructed of high-strength steel. The steering and power of the chassis can be easily controlled through the remote control, and the PIXLOOP chassis can be equipped with an autonomous driving system. This manual will use the HOOKE 4-wheel driving and 4-wheel steering version to introduce each function of the chassis in details while allowing you to quickly use this product.

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1) This manual has included various function introductions and instructions as much as possible at the time of printing. However, due to continuous improvement of product functions and design changes, there may still be discrepancies with the product you purchased. Due to product updates, there may be deviations between this manual and the actual product in terms of color, appearance, etc., please refer to the actual product.

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2) In order to protect the legitimate rights and interests of users, please be sure to read the instruction manual we provide with this product carefully before using this product. The chassis provider reserves the right to update the above documents. Please read this manual carefully before use and operate this product according to the manual.

3) 本产品不适合未满18周岁及其他不具备完全民事行为能力的人士使用，请您避免上述人士接触本产品，在有上述人士出现的场合操作时请您格外注意。

3) This product is not suitable for use by persons under the age of 18 or other persons who do not have full capacity for civil conduct. Please avoid contact with this product by the above-mentioned persons, and please pay special attention when operating in places where the above-mentioned persons appear.

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5) 在使用本产品的过程中，请您务必严格遵守并执行包括但不限于说明书里的要求。对于违反说明书所提示的使用行为或不可抗因素导致的一切人身伤害、事故、财产损失、法律纠纷，及其他一切造成利益冲突的不利事件，均由用户自己承担相关责任和损失，产品提供商将不承担任何责任。

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6) 用户使用本产品直接或间接发生的任何违反法律规定的行为，套件提供商将不承担任何责任。

6) The chassis provider will not bear any responsibility for any violation of laws and regulations that occurs directly or indirectly when the user uses this product.

II. 产品介绍Product Introduction

2.1 产品清单Product LIST



HOOKE底盘
Hooke Chassis



工业遥控器
Industrial Remote
Control



便携充电器
portable charger

工具箱×1 Toolbox×1

合格证×1 Certificate of Conformity×1

保修卡×1 Warranty Card ×1

植树证书×1 Tree Planting Certificate×1

备用零件 Spare parts

附件 Annex

注：图片仅供参考，选配版本略有不同，请以实物为准。Note: The pictures are for reference only, the optional version is slightly different, please refer to the actual product.

2.2 基本功能Basic Function

底盘通过无线遥控器与底盘本体连接，通过无线遥控器或者其他CAN通讯控制系统可以对底盘进行基本控制，也可以部署自动驾驶后实现自动驾驶功能，其中包括：

The chassis is connected to the chassis body through a wireless remote control, and the basic control of the chassis can be performed through a wireless remote control or other CAN communication control systems, and autonomous driving can also be deployed to realize autonomous driving functions, including:

四轮独立运动控制【后驱，四驱，高级功能下(选配)可开放四轮独立控制和差速转向模式】；

Four-wheel independent motion control [rear drive, four-wheel drive, four-wheel independent control and differential steering mode can be opened under advanced functions (optional)].

底盘挡位切换(进档，退档，空挡)

Chassis gear switching (forward gear, reverse gear, neutral gear).

两种转向模式(前轮阿克曼转向模式，四轮转向模式)；

Two steering modes (front wheel Ackerman steering mode, four-wheel steering mode).

底盘电子驻车控制/液压制动控制； Chassis Electric Parking Brake (EPB)/hydraulic brake control.

底盘制动控制； Chassis brake control.

多种电压平台供电功能； Multiple voltage platform power supply function.

自动驾驶/遥控驾驶切换；Autonomous driving/remote control switching;

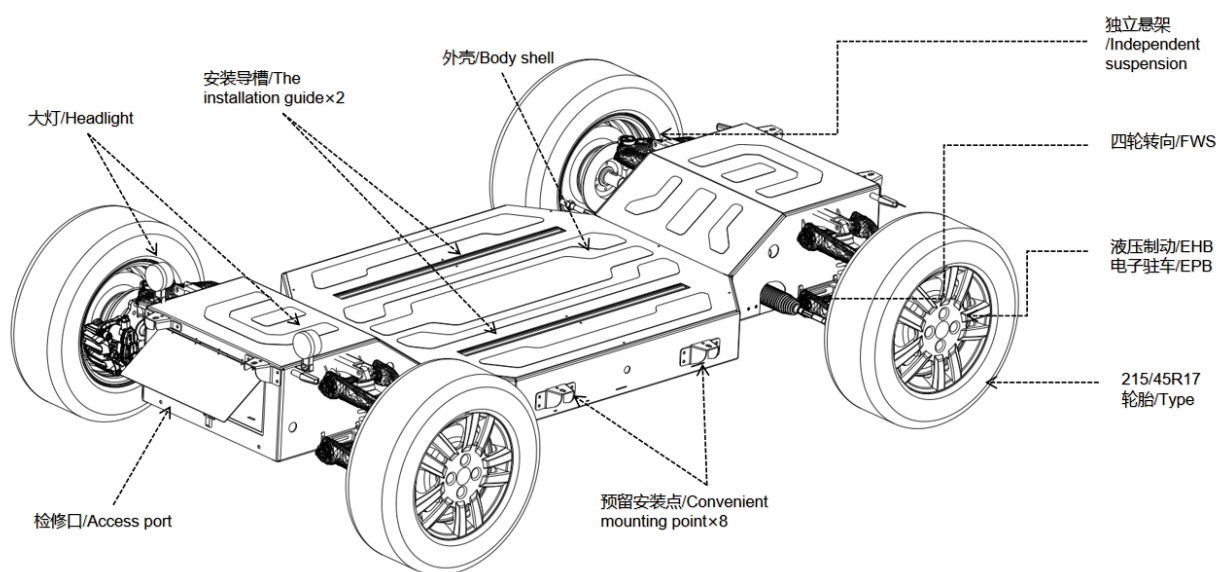
低速行车模式；Low speed driving mode.

高级自动驾驶开发等。Advanced autonomous driving development, etc.

2.3 产品概述Product Overview

PIXLOOP底盘主体承重结构采用高强度钢装配而成，表面盖板为喷涂铝板。车轮悬架部分采用专业冲压加工方式成型，设计过程采用有限元分析，充分满足车辆运动控制和不同场景因素带来的各种冲击和挑战。表面成形盖板可提供高强度机械性能还兼具美观。底盘车身采用互通箱体式结构设计，为底盘设备安装和后期扩展提供相对较大的空间冗余。

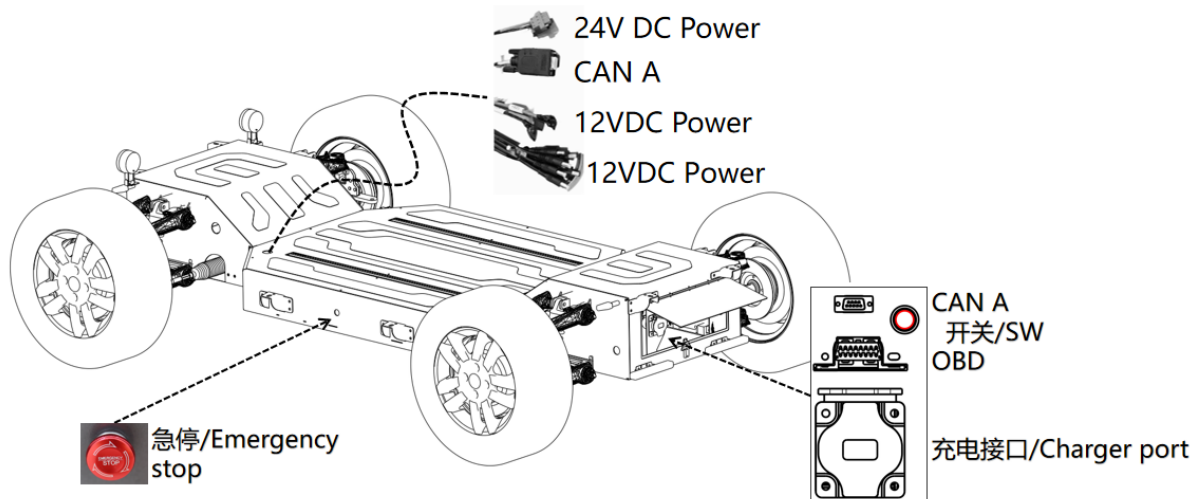
The main load-bearing structure of the PIXLOOP chassis is assembled with high-strength steel, and the surface cover is sprayed aluminum. The wheel suspension part is formed by professional stamping, and the design process uses finite element analysis to fully meet the various impacts and challenges brought about by vehicle motion control and different scene factors. Surface formed decking provides high mechanical strength combined with aesthetics. The chassis body adopts an intercommunicating box-type structure design, which provides relatively large space redundancy for chassis equipment installation and later expansion.



2.3.1 开关接口Switch And Connection Ports

底盘开关以及开放接口示意图如下：

The schematic diagram of the chassis switch and the open interface is as follows:



急停/Emergency Stop

车身急停开关按压后触发，触发立即生效，底盘失去动力，转向灯“双闪”，底盘进入驻车状态。按照图示右旋即可解除急停状态。

The emergency stop switch of the body is triggered after being pressed, and the trigger takes effect immediately. The chassis loses power, the turn signal "double flashes", and the chassis enters the parking state. Rotate right as shown in the figure to release the emergency stop state.

CAN A

自动驾驶CAN，用于接收命令和反馈底盘信息，位于车身后部检修板下和底盘表面，一共两处。

The autonomous driving CAN is used to receive commands and feedback chassis

information. It is located under the rear access panel of the chassis body and on the surface of the chassis. There are two places in total.

OBD

Fault access port for chassis inspection.

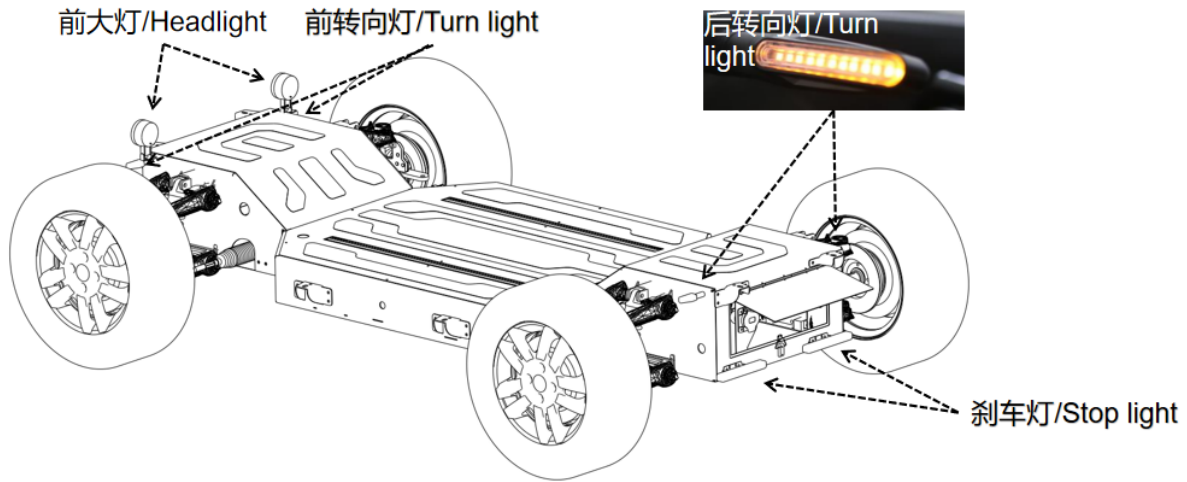
电源接口/Power Interface

24V电源接口，提供额定300W 24V直流电源；12V额定500W直流电源，提供两种类型多个12V直流电源接口。对应线束上标注有电压标识。

The 24V power interface provides a rated 300W 24V DC power supply; the 12V rated 500W DC power supply provides two types of multiple 12V DC power interfaces. The voltage mark is marked on the corresponding wiring harness.

2.3.2 灯光指示Light Indication

底盘提供了车辆基本的灯光系统，与传统车辆相比，包含前大灯，前后转向灯，制动灯
The chassis provides the basic lighting system of the vehicle. Compared with traditional vehicles, it includes headlights, front and rear turn signals, and brake lights.



前大灯/Headlight

底盘大灯由遥控器拨动开关控制。底盘打开时，当拨动开关位于下位时，大灯处于蓝色微亮状态；当拨动开关位于上位时，大灯处于常亮状态，灯光颜色白色；同时大灯提供系统开发接口，满足二次大灯控制策略开发。

The chassis headlights are controlled by the remote control toggle switch. Open the chassis, when the toggle switch is in the down position, the headlights are dimly lit in blue. When the toggle switch is in the upper position, the headlights are always on, and the light color is white. At the same time, the headlights provide a system development interface to meet the development of secondary headlight control strategies.

转向灯/Turning light

底盘转向灯为标准转向状态灯，前后左右转向灯规格相同。默认情况下转向灯为自

动控制，且同时提供系统开发接口。底盘触发急停或失联时四个转向灯持续“双闪”

The chassis turning lights are standard turning status lights, and the specifications of the front, rear, left, and right turn lights are the same. By default, the turning lights are automatically controlled, and a system development interface is provided at the same time. When the chassis triggers an emergency stop or loses contact, the four turning lights continue to "double flash".

刹车灯/Brake light(Stop light)

底盘刹车灯位于车身尾部，又叫尾灯。当系统开机后尾灯保持持续点亮状态，刹车时候高亮。同时尾灯提供系统开发接口，满足二次尾灯控制策略开发。

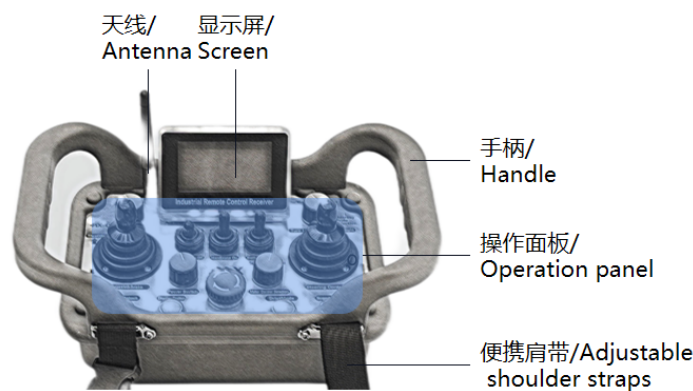
The chassis brake light is located at the rear of the body, also known as the tail light. When the system is turned on, the taillights will remain on continuously, and will be bright when braking. At the same time, the taillight provides a system development interface to meet the secondary taillight control strategy development.

2.3.3 无线遥控器Remote Control

无线遥控器是与 PIX底盘连接的基本操控器，所有车辆基本控制都能在无线遥控器运行情况下实现车辆控制，也是非车身控制之外权限最高的底盘控制接口。

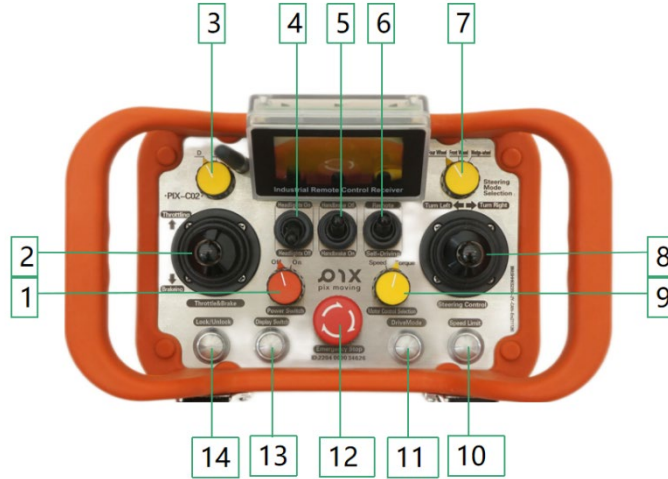
The wireless remote control is the basic controller connected to the PIX chassis. All basic vehicle controls can be realized under the condition of the wireless remote control running. It is also the chassis control interface with the highest authority other than non-body control.

遥控器频段 Remote control frequency band	Remote 443MHz	接收机通讯接口 Receiver communication interface	CAN 接口 CAN Interface
遥控器供电方式 Remote control power supply mode	3×1.5V 五号干电池 3×1.5V No.5 dry cell	遥控器显示器 Remote control display	LCD 显示屏 LCD display
接收机供电 Receiver power supply	12-24V DC 供电 12-24V DC power supply	通讯距离 Communication distance	150M (视周围环境而定) 150M (depending on the surrounding environment)



操作面板集成多种开关，分别包含遥控器摇杆×2；两段拨动开关×3；按钮×4；旋钮开关×4，功能介绍如下。

The operation panel integrates a variety of switches, including remote control joystick × 2; two-stage toggle switch × 3; button × 4; knob switch × 4, the functions are introduced as follows.



编号/No	按键名称/Button Name	编号/No	按键名称/Button Name
1	遥控开关Power switch	8	转向推杆 Steering push rod
2	加速/制动推杆 Throttling/Braking push rod	9	动力控制模式Power control mode
3	挡位切换 Gear switch: D N R	10	低速模式开关 Low speed mode switch
4	大灯开关 Headlight switch	11	四驱切换开关 Drive Mode(Four-wheel drive switch)
5	驻车开关 Parking switch	12	急停按钮 Emergency Stop
6	控制模式 Control mode: Remote Control/Autonomous driving	13	显示页面切换 Display switch
7	转向模式Steering Mode	14	遥控器锁定 Lock

电源开关Power Switch

电源开关为遥控器上电开关。

The power switch is the power-on switch of the remote control.

加速/制动推杆Throttling/Braking push rod

此摇杆控制车辆油门信号与制动信号，油门控制和制动控制都为 0-1024 级单向摇杆控制，上推该摇杆实现底盘加速操作，下推实现底盘电子制动操作。

This push rod controls the accelerator signal and brake signal of the vehicle. Both the accelerator control and the brake control are 0-1024 one-way push rod controls. Push the rod up to realize the chassis acceleration operation, and push it down to realize the chassis electronic brake operation.

挡位切换Gear switch: D N R

该旋钮控制车辆挡位切换，提供 D, N, R 三种挡位模式。The knob controls the vehicle gear switch, providing D, N, R three gear modes.

D 档：前进挡位，推动油门时车辆向前运动

D gear: Forward gear, the vehicle moves forward when the accelerator is pushed.

N 档：空挡位同时为泊车挡位，推动油门时车辆无动作响应。

N gear: The neutral gear is also the parking gear, and the vehicle does not respond when the accelerator is pushed.

R 档：倒退挡位，推动油门时车辆向后运动

R gear: Reverse gear, the vehicle moves backward when the accelerator is pushed.

驻车开关 Parking switch

为两段拨动开关，可控制电子驻车。拨动开关处于上位为关闭停机驻车功能。拨动开关处于下位为启动停机驻车功能

It is a two-stage push rod switch, which can control the electronic parking brake(EPB). When the push rod switch is in the upper position, the parking function is turned off. The push rod switch is in the down position to start the parking function

控制模式 Control mode: Remote Control/Autonomous driving

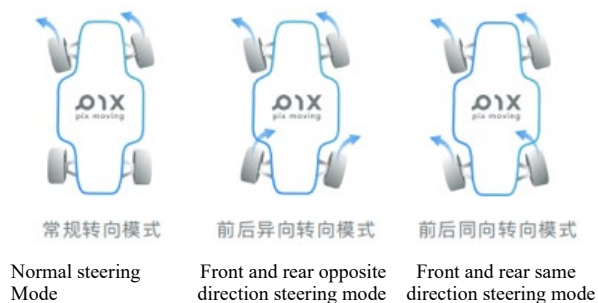
该按钮可切换底盘控制模式。往下拨按钮切换为自动驾驶模式，往上拨动按钮退出自动驾驶模式切换为人工遥控模式。

This button toggles the chassis control mode. Push the rod down to switch to the autonomous driving mode, and push the rod up to exit the autonomous driving mode and switch to the manual remote control mode.

转向模式Steering Mode

该旋钮控制车辆转向模式切换，提供提供常规转向模式、前后异向转向模式、前后同向转向模式三种转向模式。

The knob controls the switching of the steering mode of the chassis, providing three steering modes: normal steering mode, front and rear opposite direction steering mode, and front and rear same direction steering mode.



转向推杆 Steering push rod

此摇杆控制车辆转向信号，转向控制为 -1024至+1024 级双向摇杆控制。向左拨动摇杆，底盘实现左转。向右拨动摇杆，底盘实现右转。This push rod controls the steering signal of the chassis, and the steering control is a two-way push rod control with levels from -1024 to +1024. Move the push rod to the left, and the chassis will turn left. Move the push rod to the right, and the chassis will turn right.

动力控制模式Power control mode

车辆支持扭矩控制方法，扭矩控制模式下，遥控器输出的是扭矩值，即电机的驱动力的大小。The chassis supports the torque control method. In the torque control mode, the remote controller outputs the torque value, that is, the driving force of the motor.

低速模式开关 Low speed mode switch

该按钮可启动低速模式。低速模式主要是为了防止操作者在不熟悉操作方式时产生~~底盘~~速度过快发生危险的情况。在速度控制模式下，限速5km/h。油门模式下，当速度大于4km/h时，底盘会限制请求的扭矩，超过5km/h时请求扭矩会减小为0。若在下坡，由于重力作用，速度可能会超过5km/h。

This button activates low speed mode. The low-speed mode is mainly to prevent the chassis speed from being too fast and causing danger when the operator is not familiar with the operation method. In speed control mode, the speed limit is 5km/h. In throttle mode, when the speed is greater than 4km/h, the chassis will limit the requested torque, and the requested torque will be reduced to 0 when the speed exceeds 5km/h. If you are going downhill, the speed may exceed 5km/h due to gravity.

四驱切换开关 Drive Mode(Four-wheel drive switch)

电机驱动两驱和四驱切换。The switching of two-wheel motor driving and four-wheel motor driving

急停按钮 Emergency Stop

控制车辆急停。按下急停开关按钮底盘启动紧急停止，右旋急停开关按钮为释放紧急停止功能。

Control the vehicle to an emergency stop. Press the emergency stop switch button to start the emergency stop of the chassis, and turn the emergency stop switch button to the right to release the emergency stop function.
注：急停开关仅在紧急情况下使用。请勿将急停开关按钮长期处于按下状态。

NOTE: The emergency stop switch is for emergency use only. Do not keep the emergency stop switch button pressed for a long time.

显示页面切换 Display Switch

遥控器主要有两个页面，可以通过遥控上Display Switch切换。第一显示屏主要显示操作指令和车辆基本状态，第二显示界面主要显示车辆里程、系统具体故障和遥控器信号强度（信号强度由0-255代表由弱到强）。

The remote control mainly has two pages, which can be switched through the Display Switch on the remote control. The first display screen mainly displays the operation instructions and the basic status of the chassis, and the second display interface mainly displays the chassis mileage, specific system faults and the signal strength of the remote control (the signal strength ranges from 0-255 to represent from weak to strong).

遥控器锁定 Lock

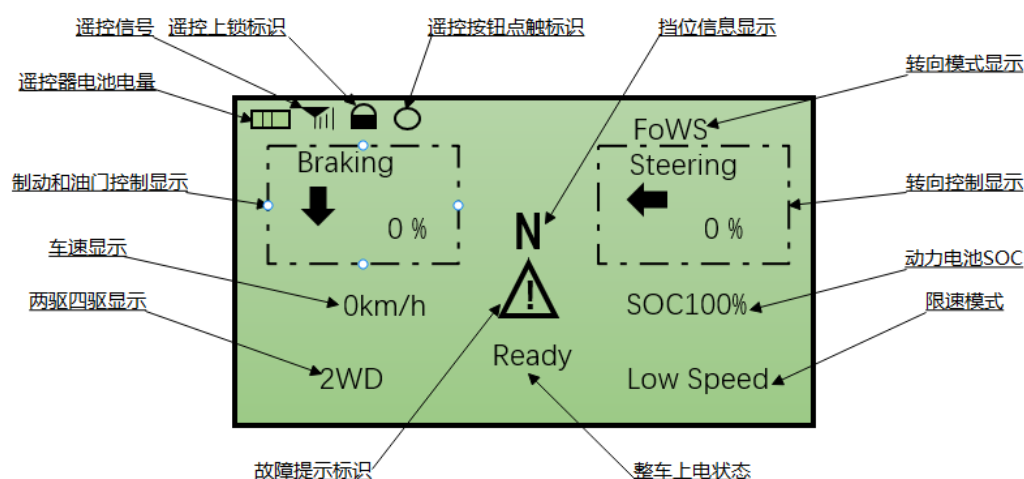
遥控器锁定后不可操作。

The remote control cannot be operated after it is locked.

显示面板介绍 Display panel introduction

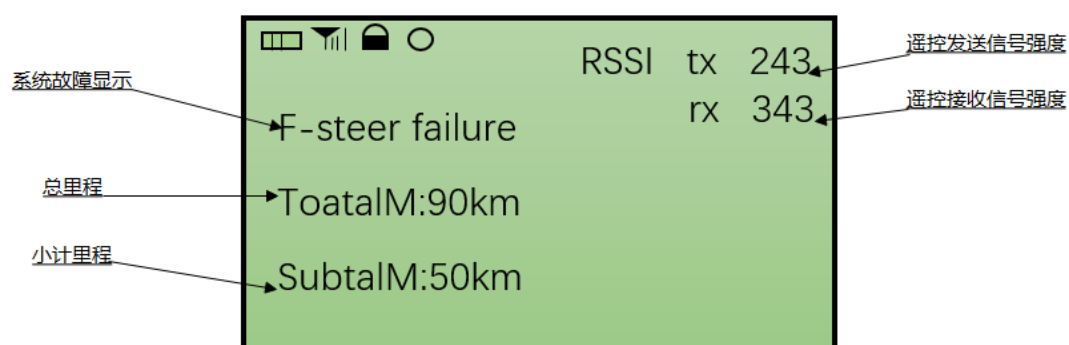
遥控器有两个主页面，可以通过遥控上Display Switch切换。第一显示屏主要显示操作指令和车辆基本状态，详细的介绍如下图。

The remote control has two main pages, which can be switched through the Display Switch on the remote control. The first display screen mainly displays the operation instructions and the basic status of the chassis. The detailed introduction is as shown in the figure below.



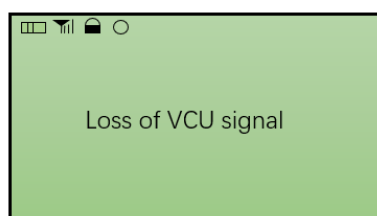
第二显示界面主要显示车辆里程、系统具体故障和遥控器信号强度，详细的介绍如下图（信号强度由0-255代表由弱到强）。

The second display interface mainly displays the chassis mileage, specific system faults and the signal strength of the remote control. The detailed introduction is shown in the figure below (the signal strength ranges from 0-255 to represent from weak to strong).



此外，遥控器还有提示界面，用于遥控器的故障信号显示。

In addition, the remote control has a prompt interface for displaying fault signals of the remote control.



Loss of VCU signal-VCU信号丢失
Push rod back to center-推杆故障
Low battery-遥控器电池电量低
Lost connection-无线通讯异常

遥控器安全操作提示Tips for safe operation of the remote control

- ◆ 遥控器开启和关闭时需保证各键位处于关闭状态，即驻车、空挡、遥控接管，无油门；When the remote control is turned on and off, it is necessary to ensure that each key is in the closed state, that is, parking, neutral, remote control takeover, and no throttle.
- ◆ 进入遥控模式后，请勿一次性将油门推到最大，以免发生危险，应慢慢推动油门控制方向验证各键位是否正常工作；After entering the remote control mode, do not push the accelerator to the maximum at one time to avoid danger, but slowly push the accelerator to control the direction to verify whether the buttons are working normally.
- ◆ 在空旷环境遥控器使用范围保证在50米；The range of the remote control is guaranteed to be 50 meters in an open environment
- ◆ 停车后保证车辆驻车、空挡状态，避免在坡道停车；After parking, ensure that the vehicle is parked and in neutral, and avoid parking on ramps.
- ◆ 出于安全考虑，任何人在使用遥控器前需仔细阅读遥控器相关说明，切勿将遥控器交给不熟悉相关操作的人员使用，以免发生危险；For safety reasons, anyone should carefully read the relevant instructions of the remote control before using it, and do not hand over the remote control to people who are not familiar with the relevant operations to avoid danger;
- ◆ 遥控器信号失联后底盘触发安全保护机制，急停驻车；After the remote control signal is lost, the chassis triggers the safety protection mechanism, and the chassis stops in an emergency
- ◆ 为了您的安全，在进入自动驾驶模式前，请务必提前熟悉遥控器的各项操作，尤其关于急停功能的使用，请在每次进入自动驾驶模式前进行遥控操作确认。为了您的安全，我们已在工控机与底层协议层定义了遥控器的绝对优先权，即平台移动的任何状态下（请在每次运行前确保遥控器电量充足），只要遥控器上电且处于遥控器接管状态，平台即进入人工接管模式，为您的安全测试保驾护航。

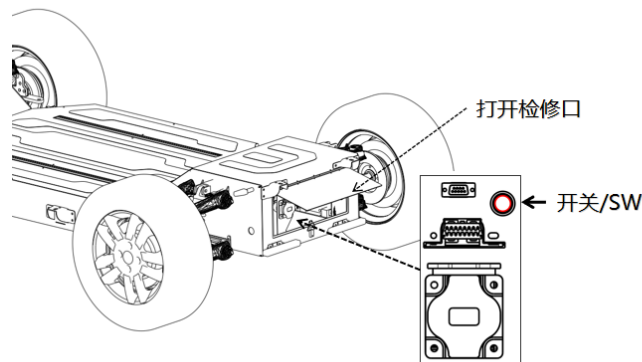
For your safety, before entering the automatic driving mode, please be sure to be familiar with the various operations of the remote control in advance, especially for the use of the emergency stop function, please confirm the remote control operation before entering the autonomous driving mode each time. For your safety, we have defined the absolute priority of the remote control at the industrial computer and the underlying protocol layer, that is, in any state of platform movement (please ensure that the remote control has sufficient power before each operation), as long as the remote control is powered on and in the remote control takeover state, the platform will enter the manual takeover mode to escort your safety test.

III. 使用与开发 Usage and development

3.1 遥控底盘 Remote Control Chassis

底盘开机 Chassis power on

- 打开车身尾部检修盖板 Open the rear inspection cover of the chassis body
- 按下底盘电源开关，底盘进入初始化和自检，大约需要5S，自检完成后启动。
Press the chassis power switch, the chassis enters initialization and self-test, it takes about 5 seconds, and starts after the self-test is completed.



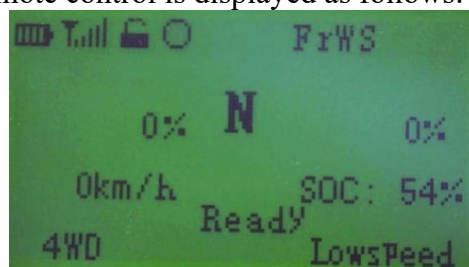
提示：开关自带指示灯，开关打开指示灯呈红色，底盘关闭指示灯熄灭。

Tip: The switch has its own indicator light. The switch open light is red and the chassis close light is off.

连接遥控器 Connect the remote control

旋转无线遥控器电源开关至ON档，开启遥控器开关，并处于遥控接管模式，**拨杆处于驻车、空挡，左右摇杆处于中间位置状态，使遥控器上电，进入遥控接管模式。**启动成功后遥控器页面显示如下：

Rotate the power switch of the wireless remote control to the ON position, turn on the switch of the remote control, and it is in the remote control takeover mode. The button is in the park and neutral positions, and the left and right push rods are in the middle position, so that the remote control is powered on and enters the remote control takeover mode. After the startup is successful, the page of the remote control is displayed as follows:



提示：如果出现异常，请检查车身急停和遥控器按钮，参照常见问题处理，确认无误后再进行下一步。

Tips: If there is an abnormality, please check the emergency stop and the remote control button, refer to the common problems to deal with, and proceed to the next step after confirming that there is no error

底盘关机 Chassis shutdown

确保车辆停放正确后，将遥控器的驻车按钮启动，关闭遥控器电源按钮，按下底盘电源按键，确保车辆下电。

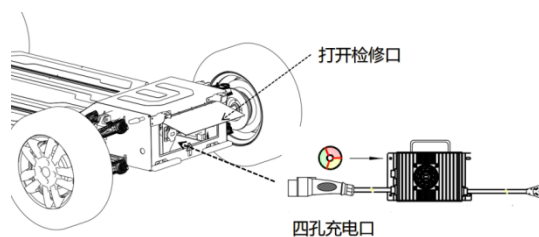
After ensuring that the vehicle is parked correctly, activate the parking button of the remote control, turn off the power button of the remote control, and press the power button of the chassis to ensure that the vehicle is powered off.

3.2 充电Charging

充电环境应保持干燥通风，车辆及充电设备周围2米范围内不能有可燃物,按照以下步骤充电：

The charging environment should be kept dry and ventilated. There should be no combustibles within 2 meters around the chassis and charging equipment. Follow the steps below to charge

- 关闭底盘电源 Power off the chassis
- 连接便携式充电机与底盘充电插座
Connecting the Portable Charger to the Chassis Charging Socket
- 连接交流电源(中国电压是220V) Connect to AC power (The voltage standard in China is 220V)



- 指示灯的状态 Status of the indicator light
 - 红色和绿色交替闪烁（间隔一秒），电池没有连接，未能检测到电池；
Red and green flash alternately (one second interval), the battery is not connected, and the battery cannot be detected;
 - 红色指示灯每秒闪烁一次 风机转动， 电池容量 < 80%；
The red indicator light flashes once per second, the fan is running, and the battery capacity is <80%;
 - 黄灯/（绿灯）指示灯每一秒闪烁一次 电池电量≥80%

The yellow light/(green light) indicator flashes once every second, and the battery power is $\geq 80\%$

- 绿色指示灯常亮（关机）电池电量= 100%

Green light is always on (power off) battery power = 100%

- 当充电机指示灯为绿灯常亮说明充电完成；
When the indicator light of the charger is green, it means that the charging is completed;
- 拔下充电机电源接头； Unplug the power connector of the charger
- 拔下充电机充电口接头。 Unplug the charging port connector of the charger.

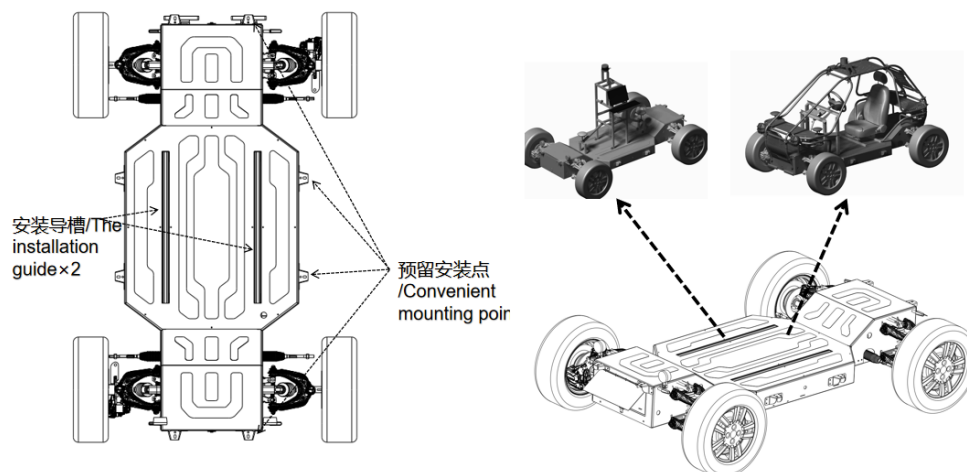
3.3 开发使用Development Use

3.3.1 快速搭载自定义上装Upper Pod Development Support

预留安装点 Reserved mounting point

硬件安装点，底盘提供便捷的硬件安装点，客户可以依据布置的安装点快速简易的安装支架和设备，示例如下：

Hardware mount points. The chassis provides convenient hardware installation points, and customers can quickly and easily install brackets and equipment according to the arranged installation points, examples are as follows:



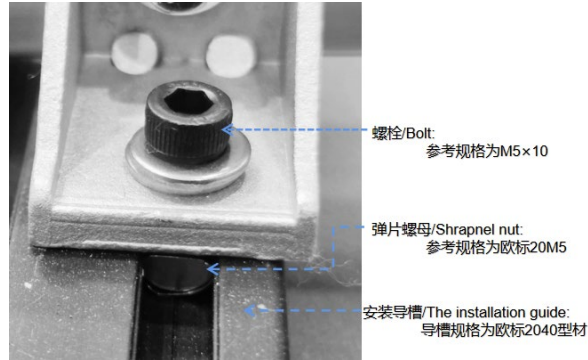
连接推荐采用以下规格安装连接材料：

弹片螺母/Shrapnel nut，参考规格为欧标20M5，安装导槽/The installation guide，导槽规格为欧标2040型材，螺栓/Bolt，参考规格为M5×10

The connection is recommended to use the following specifications to install the connection material:

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Shrapnel nut, the reference specification of which is European standard 20M5. The installation guide, the specification of which is the European standard 2040 profile. Bolt, the reference specification of which is M5×10.

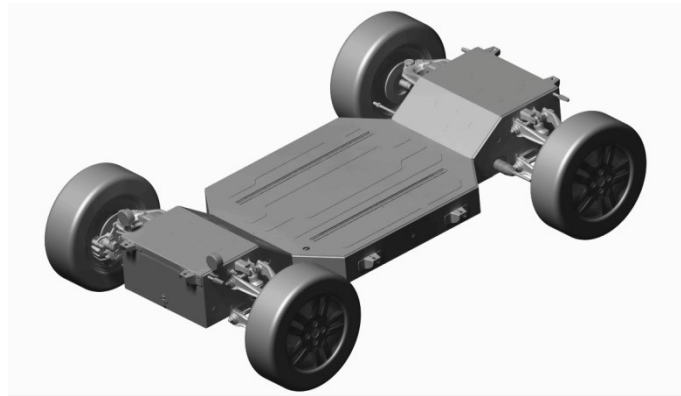


底盘边界模型 Chassis boundary model

模型包含cgr和3dxml两个格式，是消参格式，客户可以用目前主流的3D工程软件打开，包括不限于达索CATIA、UG、Autodesk。CGR 在CATIA打开需要以插入形式打开。注意此文件不可随意传播，部分部件和实物有差异，仅可作为参考。

The model contains two formats of cgr and 3dxml, which are deparameter formats, and customers can open them with current mainstream 3D engineering software, including but not limited to Dassault CATIA, UG, and Autodesk. CGR needs to be opened in insert mode when opening in CATIA.

Note that this file cannot be distributed at will, and some parts and objects are different, so it can only be used as a reference.



3.3.2 自动驾驶应用Autonomous Driving Application

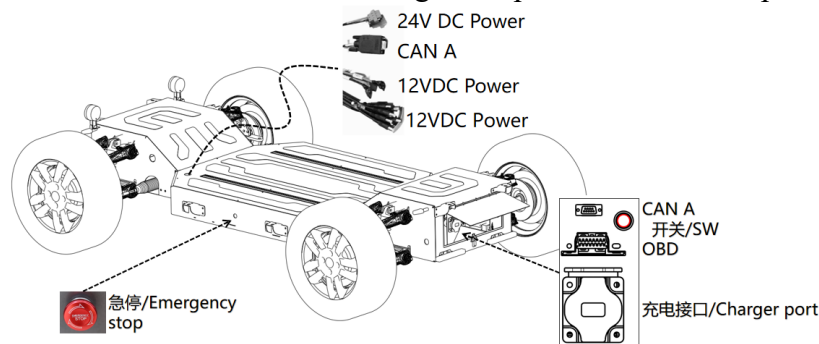
底盘为自动驾驶实施提供自动驾驶常用电源，专用CAN接口，标准底盘开放自动驾驶协议和对应DBC文件。The chassis provides common power supply for autonomous driving, dedicated CAN interface, standard chassis open automatic driving protocol and corresponding DBC files.

- ◆ 电源接口输出额定功率累计值为12V500W, 24V300W, 接口类型参考“产品概述—开关接口”。

The cumulative output rated power of the power interface is 12V500W, 24V300W, and the interface type refers to "Product Overview-Switch Interface".

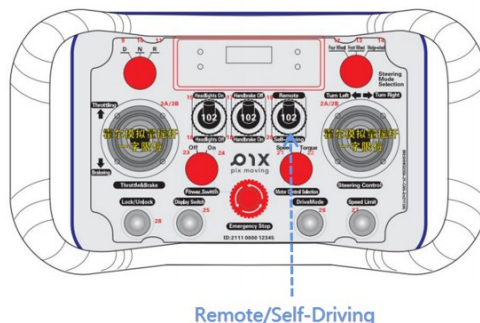
- ◆ 提供两个自动驾驶专用CAN接口, 插件型号DB9, CAN2.0B通讯, 属于底盘CAN-A
Provide two dedicated CAN interfaces for autonomous driving, plug-in model DB9, CAN2.0B communication, belonging to the chassis CAN-A
- ◆ 提供标准底盘自动驾驶CAN协议和对应DBC文件。

Provide standard chassis autonomous driving CAN protocol and corresponding DBC files.



打开自动驾驶时, 需要在遥控上切换自动驾驶模式。

When turning on the autonomous driving function, you need to switch the autonomous driving mode on the remote control.



3.3.3 高级权限（选配）Advanced Rights (Optional)

支持分布式驱动研发, 实现高级运动控制, 灯光控制接口开放, 高级驾驶接口是为需要研究底盘动力学模型和需要根据自己的实际应用去优化动力学模型的控制接口。启用高级模式后, 可以绕过VCU的主要控制算法和逻辑, 直接下达指令到各个运动执行机构。此项需要在采购时下单开放权限, 常规车辆不具备此功能。

Support distributed drive research and development, realize advanced motion control, and open lighting control interface. The advanced driving interface is a control interface for studying the chassis dynamics model and optimizing the dynamics model according to your actual application. After the advanced mode is enabled, the main control algorithm and logic of the

VCU can be bypassed, and instructions can be issued directly to each motion actuator. This item needs to be opened when purchasing an order, and conventional vehicles do not have this function.

IV. 车辆安全机制Vehicle Safety Mechanism

4.1 车辆遥控防误触Prevent The Wrong Touch

为保证车辆行驶安全，遥控器加速推杆设置误操作机制，即在猛推加速推杆时底盘不动作，防止车辆急速“窜出”。

In order to ensure the driving safety of the vehicle, the accelerator push rod of the remote control is equipped with a misoperation mechanism, that is, the chassis does not move when the accelerator pusher is pushed violently, preventing the vehicle from "jumping out" rapidly.

注意：安全起见，遥控驾驶车辆时车辆请确保车辆前方和后方10M内无行人和障碍物，并时刻注意危险情况，随时做好接管准备，正常测试建议遥控器专人负责。

Note: For the sake of safety, please ensure that there are no pedestrians and obstacles within 10M in front and behind the chassis when driving the vehicle remotely, and always pay attention to dangerous situations, and be ready to take over at any time. It is recommended that the remote control be responsible for normal testing.

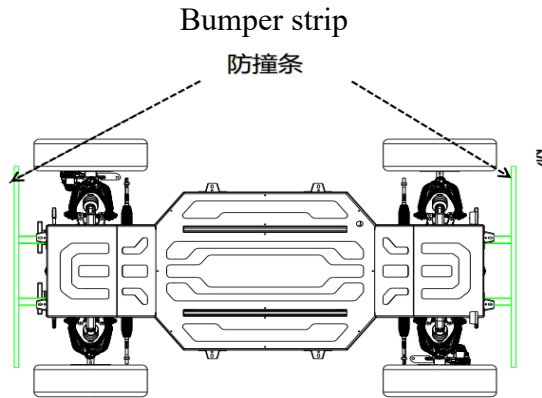
4.2 碰撞传感器（选配）Collision Sensor (Optional)

标准底盘在车体前舱和后舱标配预留了防撞条**线束**接口。

选配防撞条的车辆前后分别有一条防撞条，能够感应防撞条与车体间的压力从而判断是否发生了碰撞，碰撞传感器最小触发力 $\leq 50\text{N}$ ，触发后车辆立即进入紧急制动状态，车辆进入紧急制动模式，车辆会以最大刹车力进行制动，此模式下车辆无法移动，且无法控制。车辆与碰撞物体脱离接触或者防撞条形变恢复（内部开关断开连接），可解除紧急制动状态。

The standard chassis has reserved anti-collision strip wire harness interfaces in the front compartment and rear compartment of the chassis body.

Chassis equipped with anti-collision strips has an anti-collision strip at the front and rear respectively, which can sense the pressure between the anti-collision strip and the chassis body to determine whether a collision has occurred. The minimum trigger force of the collision sensor is $\leq 50\text{N}$, and the chassis immediately enters the emergency braking state after triggering, the vehicle enters the emergency braking mode, and the chassis will brake with the maximum braking force. In this mode, the vehicle cannot move and cannot be controlled. The emergency braking state can be released when the chassis loses contact with the collision object or the deformation of the anti-collision strip recovers (the internal switch is disconnected).



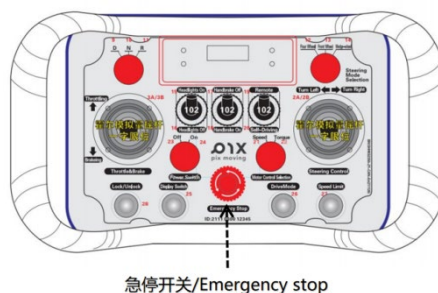
4.3 遥控器急停Remote Control Emergency Stop

- 当车辆遇到紧急情况时按下遥控器中间急停按钮启动底盘紧急停止，驻车制动，遥控器界面将会显示“ESTOP!”。

When the chassis encounters an emergency, press the emergency stop button in the middle of the remote control to start the emergency stop of the chassis and the parking brake, and the remote control interface will display "ESTOP!".

- 右旋急停按钮为释放紧急停止功能。急停后车辆转向灯双闪，尾灯强亮，解除急停恢复。

Turn the emergency stop button to the right to release the emergency stop function. After the emergency stop, the chassis' turn signal flashes twice, and the tail light is brightly lit, and the emergency stop is released and resumed.



4.4 车身急停开关Body Emergency Stop Switch

急停开关作用是给车辆提供紧急停车信号，该按键位于车身左侧。

The function of the emergency stop switch is to provide an emergency stop signal to the vehicle, and the button is located on the left side of the chassis body.

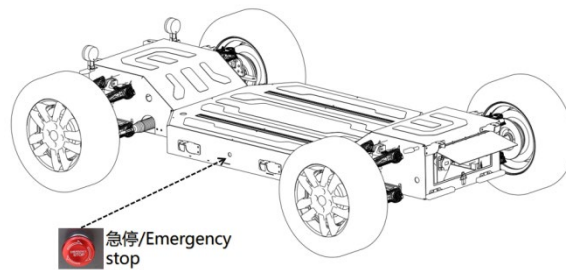
- 按下车身急停开关，触发紧急制动，车辆进入紧急制动状态，车辆会以最大刹车力进行制动。

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Press the emergency stop switch on the body to trigger emergency braking, the chassis enters the emergency braking state, and the chassis will brake with the maximum braking force.

- 急停后车辆转向灯双闪，尾灯强亮，解除急停恢复。在紧急制动状态下，车辆挡位为空挡，油门驱动无效。将急停按钮顺时针旋起即解除紧急制动状态。

After the emergency stop, the chassis 'turning lights flash twice, and the tail light is brightly lit, and the emergency stop is released and resumed. In the state of emergency braking, the gear of the chassis is neutral, and the accelerator drive is invalid. Turn the emergency stop button clockwise to release the emergency braking state.



V. 底盘性能参数Performance Parameter of Chassis

5.1 车身整体参数The overall parameters of the chassis

外形尺寸 (mm)	2510×1700×646
Dimensions (mm)	
轴距 (mm)	1900
Wheelbase (mm)	
前/后轮距 (mm)	1465/1465
Front/Rear Wheel Track (mm)	
前/后悬 (mm)	237/237
Front/Rear Overhang (mm)	
最小离地间隙 (mm) (满载)	175/170
Min Ground Clearance (mm) (Full Load)	
最小转弯半径 (m)	3/6
Min Turning Radius (m)	
整备质量 (kg)	450/420
Curb Weight (kg)	
最大允许载质量 (kg)	500/400
Max Allowable Load Mass (kg)	
满载质量 (kg)	950/820
Full Load Mass (kg)	
常规车速 (Km/h)	≤40
Normal Speed (Km/h)	
最高车速 (Km/h) 空载	60/55
Max Speed (Km/h) No Load (A Disclaimer Agreement Needs to be Signed)	
最大爬坡度 (%) 空载	30/20
Max Gradability (%) No Load	
加速时间 (s) (0-30Km) 空载	4.5/9
Acceleration Time (s) (0-30Km) No Load	
内/外轮转角 (°)	30/27
Inner/Outer Wheel Steering Angle (°)	
悬挂类型 Suspension Type	前后双叉臂独立悬挂Front & Rear Double Wishbone Independent Suspension
防护等级 IP Level	IP56
悬架弹簧行程 (mm) 满载	30±3
Suspension Spring Stroke (mm) Full Load	

车架材质Frame Material	高强度钢High Strength Steel
驱动形式Drive Style	四轮驱动Four-Wheel Drive
续航里程 (Km) 空载	70
Cruising Range (Km) No Load	
5.2 转向系统参数Steering System Parameters	
转向系统主要由转向供电系统，转向机总成和转向机控制器组成。转向机供电由12VDC 标准车载系统提供，转向机和控制器均符合量产车规级标准。	
The steering system is mainly composed of the steering power supply system, the steering gear assembly and the steering gear controller. The power supply of the steering gear is provided by the 12V DC standard on-board system, and the steering gear and the controller meet the standards of mass-produced vehicles.	
额定功率 (KW)	0.5
Rated Power (KW)	
额定扭矩 (N.M)	4.5
Rated Torque (N.M)	
额定转速 (r/min)	1050
Rated Rotating Speed (r/min)	
工作电压 (V)	10.8~14
Operating Voltage (V)	
方向机转向范围 (°)	±450
Steering Range (°)	
转向结构	线控电动转向
Steering Structure	Drive-by-wire Electric Steering
转向类型	阿克曼转向
Steering Type	Ackerman Steering
转向形式	四轮/楔形转向
Steering Mode	Four-wheel/Wedge Steering

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工作温度 (°C)	-40~105	行车制动类型	线控液压制动
Operating Temperature (°C)		Braking Type for Driving	Drive-by-wire Hydraulic Braking
通讯速度 (kps)	500	驻车制动类型	电子驻车系统
Communication Speed (kps)		Braking Type for Parking	Electronic Parking System
转向精度 (°)	1	超调量 (Mpa)	0.5
Steering Accuracy (°)		Overshoot (Mpa)	
响应时间 (ms)	≤150	制动器型式	盘式
Response Time (ms)		Brake Type	Disc
转向拉杆最大输出力 (N)	10000	电机类型	永磁无刷电机
Max Output Force of Steering Rod (N)		Motor Type	Permanent Magnet Brushless Motor
通讯协议	CAN2.0B	额定功率 (KW)	0.45
Communication Protocol		Rated Power (KW)	
稳态误差 (°)	±1	额定扭矩 (N.M)	2.3
Steady-state error (°)		Rated Torque (N.M)	
超调量 (°)	≤15	额定转速 (r/min)	1700
Overshoot (°)		Rated Rotating Speed (r/min)	
控制方式	目标方向盘转角和角速度	工作电压 (V)	9~14
Control Method	Target Steering Wheel Angle & Angular Velocity	Operating Voltage (V)	
前束 (mm)	5	响应时间 (ms)	≤200
Toe-in (mm)		Response Time (ms)	
		通讯协议	CAN2.0B
		Communication Protocol	
		最大制动压力 (Mpa)	7
		Max Braking Pressure (Mpa)	
		最大压力建压时间 (ms)	≤300ms
		Max Pressure Build-up Time (ms)	
		控制方式	目标压力闭环
		Control Method	
		控制精度 (Mpa)	0.1
		Control precision (Mpa)	
		稳态误差 (Mpa)	0.3
		Steady-state Error (Mpa)	
		工作温度 (°C)	-40 ~ 85
		Operating Temperature (°C)	
		最大制动距离 (m)	≤6
		(满载@30km/h)	
		Max Braking Distance (m)	
		Full load@30km/h	

5.3制动系统参数Braking System Parameters

制动系统和传统的汽车制动系统相比，减少了人工制动相关执行机构，全车制动均采用电子控制方式完成。制动系统主要组成部分包括：制动供电系统，制动液压油管回路，刹车碟与电子卡钳。除了液压制动，整车还提供电子驻车功能。

Compared with the traditional automobile braking system, the braking system of the chassis reduces the number of actuators related to manual braking, and the braking of the whole chassis is completed by electronic control. The main components of the brake system include: brake power supply system, brake hydraulic oil pipe circuit, brake disc and electronic caliper. In addition to hydraulic braking, the vehicle also provides electronic parking functions.

行车制动类型	线控液压制动
Braking Type for Driving	Drive-by-wire Hydraulic Braking
驻车制动类型	电子驻车系统
Braking Type for Parking	Electronic Parking System
超调量 (Mpa)	0.5
Overshoot (Mpa)	
制动器型式	盘式
Brake Type	Disc
电机类型	永磁无刷电机
Motor Type	Permanent Magnet Brushless Motor
额定功率 (KW)	0.45
Rated Power (KW)	
额定扭矩 (N.M)	2.3
Rated Torque (N.M)	
额定转速 (r/min)	1700
Rated Rotating Speed (r/min)	
工作电压 (V)	9~14
Operating Voltage (V)	
响应时间 (ms)	≤200
Response Time (ms)	
通讯协议	CAN2.0B
Communication Protocol	
最大制动压力 (Mpa)	7
Max Braking Pressure (Mpa)	
最大压力建压时间 (ms)	≤300ms
Max Pressure Build-up Time (ms)	
控制方式	目标压力闭环
Control Method	
控制精度 (Mpa)	0.1
Control precision (Mpa)	
稳态误差 (Mpa)	0.3
Steady-state Error (Mpa)	
工作温度 (°C)	-40 ~ 85
Operating Temperature (°C)	
最大制动距离 (m)	≤6
(满载@30km/h)	
Max Braking Distance (m)	
Full load@30km/h	

5.4动力系统参数Power System Parameters

动力系统主要由 72VDC 供电电池，动力电池配电箱，轮毂电机，轮毂电机控制器，整车控制系统与连接线束组成。动力系统参数如下。

The power system is mainly composed of 72V DC power supply battery, power battery distribution box, hub motor, hub motor controller, vehicle control system and connecting wiring harness. The power system parameters are as follows.

额定功率 (KW)	4×3/2×3
Rated Power (KW)	
额定扭矩 (N.M)	4×126/2×126
Rated Torque (N.M)	
额定转速 (r/min)	230
Rated Rotating Speed (r/min)	
峰值功率 (KW)	4×3.8/2×3.8
Peak Power (KW)	
峰值扭矩 (N.M)	4×180
Peak Torque (N.M)	
峰值转速 (r/min)	550
Peak Rotating Speed (r/min)	
额定电压 (V)	72V
Rated Voltage (V)	
冷却方式	风冷
Cooling Method	
控速精度 (km/h)	0.1
Speed Control Accuracy (km/h)	
电机尺寸 (mm)	ø280
Motor Size (mm)	
响应速度 (ms)	≤200ms
Responding Speed (ms)	
稳态误差 (km/h)	0.2
Steady-state Error (km/h)	
通讯协议	CAN2.0B
Communication Protocol	
驱动电机类型	永磁直流无刷电机
Drive motor type	Permanent Magnet Brushless DC Motor
超调量 (km/h)	0.4
Overshoot (km/h)	

控制方式
Control Method

加速踏板开度(扭矩控制、转速控制) Accelerator Pedal Opening (Torque Control & Speed Control)

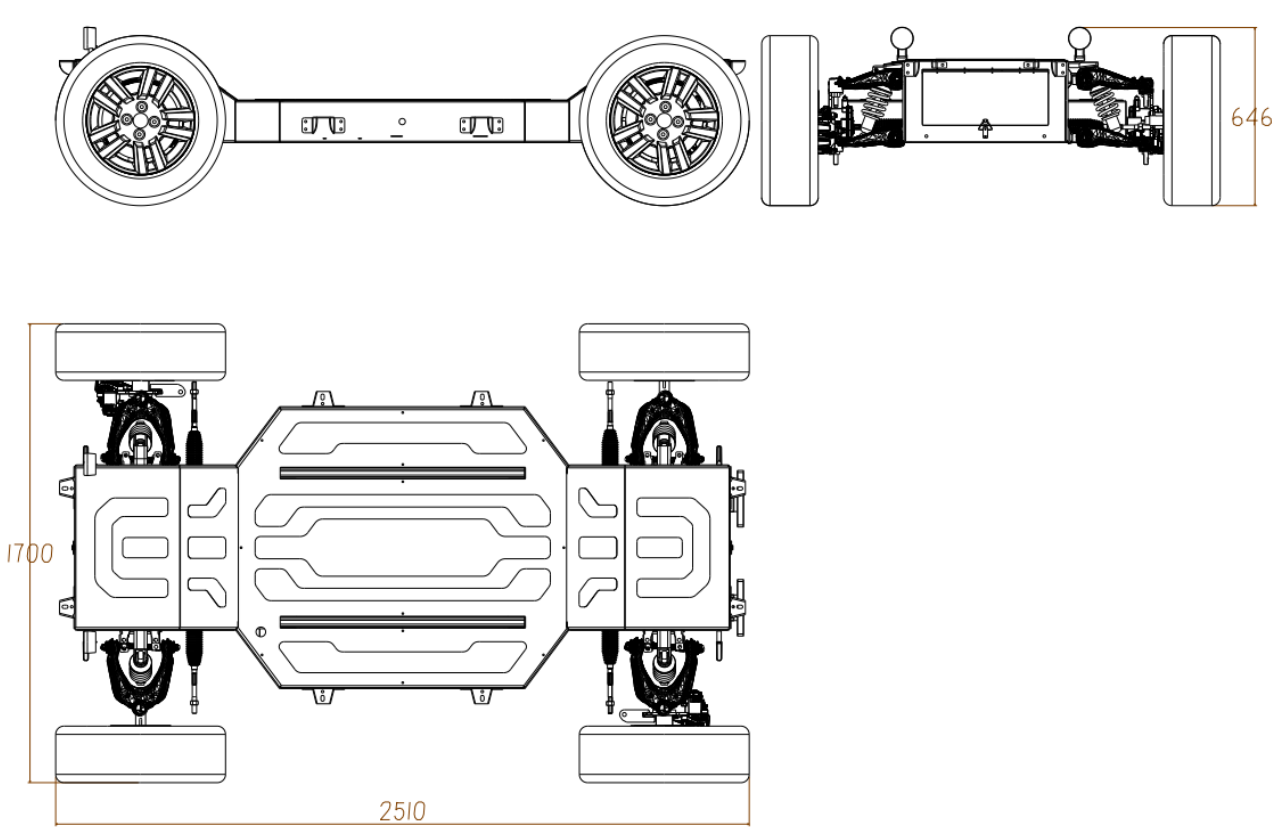
5.5供电系统参数Power Supply System Parameters

全车动力系统供电均由72V动力电池提供，同时还为12VDC 设备供电提供实时电压转换供电，整车平台除DC/DC和动力系统其他部分设备均采用12V 电压平台供电，12V 电压平台为本车基本设备供电平台

The power supply for the power system of the whole vehicle is provided by the 72V power battery. At the same time, it also provides real-time voltage conversion power supply for the power supply of 12VDC equipment. The equipment of the whole vehicle platform except DC/DC and other parts of the power system are powered by the 12V voltage platform. The 12V voltage platform is the vehicle Basic equipment power supply platform.

动力电池类型	三元锂电
Power Battery Type	Ternary Lithium Battery
电池系统额定电压 (V)	72
Rated Voltage of Battery System (V)	
电池系统额定容量 (AH)	100/150
Rated Capacity of Battery System (AH)	
电池系统能量 (kW.h)	7.2/10.8
Battery System Energy (kW.h)	
电池系统最大输出功率 (KW) (≤60S) Max Output Power of Battery System(KW)(less than or equal to 60S)	25
动力电池过放保护方式	BMS
Over-discharge Protection Method for Power Battery	
动力电池过充保护方式	BMS
Overcharge Protection Method for Power Battery	
充电方式	交流慢充
Charging Method	AC Slow Charge
充电时间 (h)	6
Charging Time (h)	
冷却方式	自然冷却
Cooling Method	Natural Cooling
低压启动电源	12V铅酸蓄电池
Low Voltage Starting Power	12V Lead-acid Battery

5.6 外形尺寸图 Dimension Drawing



VI. 常见问题处理FAQ

本节整理了各种底盘常见问题处理方法，若根据此文档未能解决，请联系相关技术人员
This section sorts out the solutions to various chassis common problems. If you can't solve it according to this document, please contact the relevant technical personnel.

Q：底盘无法驱动，遥控器无任何报警。

The chassis cannot be driven, and the remote control has no alarm.

A：第一步：检查动力电池电量是否充足；第二步：检查所有急停开关是否弹出；第三步：确认手刹是否关闭；第四步：确认是否在遥控模式下；第五步：确认刹车是否回位正常，检查车辆是否能推动；第六步：检查电机控制器高压供电是否正常；第七步：检查保险盒里电机控制器唤醒电源继电器是否正常。

Step 1: Check if the power battery is sufficient. Step 2: Check if all emergency stop switches pop up. Step 3: Check if the handbrake is off. Step 4: Check if it is in remote control mode. Step 5: Check if the brakes are on The return position is normal, check whether the vehicle can be pushed. Step 6: Check whether the high-voltage power supply of the motor controller is normal. Step 7: Check whether the wake-up power relay of the motor controller in the fuse box is normal.

Q：底盘无法开机。 Chassis cannot be powered on

A：第一步：检查所有急停开关是否弹出；第二步：检查12V蓄电池是否亏电；第三步：检查各个保险盒保险丝是否正常；第四步：检查低压保险盒是否有电；第五步：检查VCU供电是否正常；第六步：检查CAN总线是否存在短路、断路。

Step 1: Check if all emergency stop switches pop out. Step 2: Check whether the 12V battery is out of power. Step 3: Check whether the fuses of each fuse box are normal. Step 4: Check whether the low-voltage fuse box is powered. Step 5: Check whether the VCU power supply is normal. Step 6: Check whether there is a short circuit or open circuit on the CAN bus.

Q：遥控器显示SOC=0%。 The remote control shows SOC=0%

A：第一步：查看遥控器是否有其他报警，若有，对照附录1《遥控器故障码分析及处理建议》处理；第二步：检查底盘动力电池电量，给底盘充电；第三步：检查12V蓄电池是否亏电，亏电会导致动力电池无法输出，建议给蓄电池补电；第四步：检查各个保险盒保险丝是否正常，线路是否有短路、断路；第五步：读取底盘CAN报文，查看是否能读取到782报文，联系技术协助处理。

Step 1: Check if the remote controller has other alarms. If so, refer to Appendix 1 "Remote Controller Fault Code Analysis and Suggestions for Handling". Step 2: Check the power of the

chassis power battery and charge the chassis. Step 3: Check whether the 12V battery is short of electricity, which will cause the power battery to fail to output, and it is recommended to recharge the battery. Step 4: Check whether the fuses of each fuse box are normal, whether there is a short circuit or an open circuit in the line. Step 5: Read the chassis CAN message, check whether the 782 message can be read, and contact technical assistance.

Q: 底盘无法充电。 The chassis cannot be charged.

A: 第一步：确认充电顺序，先插充电枪，再插220V市电；第二步：检查总开关（旋钮开关，若有）是否打开，总开关关闭无法充电；第三步：查看充电器指示灯状态，确认充电状态（红灯慢闪表示充电中，绿灯常亮表示充电完成）；第四步：检查充电线路，是否有短路、断路；第五步：读取底盘报文，查看是否能读取到782报文，联系技术协助处理。

Step 1: Confirm the charging sequence, plug in the charging cable first, and then plug in the AC plug(The voltage standard in China is 220V) . Step 2: Check whether the main switch (knob switch, if any) is turned on, and it cannot be charged when the main switch is turned off. Step 3: Check the status of the indicator light of the charger to confirm the charging status (The red light flashes slowly to indicate charging, and the green light is always on to indicate that the charging is complete). Step 4: Check the charging circuit, whether there is a short circuit or open circuit; Step 5: Read the chassis message, check whether the 782 message can be read, and contact technical assistance to deal with it.

Q: 底盘转向失效。 Chassis steering fails.

A: 第一步：检查遥控器是否有报警，按照对应故障码处理；第二步：检查手刹、制动是否正常，确认低压供电正常；第三步：检查12V蓄电池是否亏电；第四步：检查保险盒，确认是否有保险丝烧断；第五步：检查转向电机继电器，确认是否烧坏，可交叉验证。

Step 1: Check whether the remote controller has an alarm, and deal with it according to the corresponding fault code. Step 2: Check whether the handbrake and brake are normal, and confirm that the low-voltage power supply is normal. Step 3: Check whether the 12V battery is out of power. Step 4: Check In the fuse box, confirm whether the fuse is blown. Step 5: Check the steering motor relay to confirm whether it is burnt out, which can be cross-validated.

Q: 底盘制动失效 Chassis brake failure

A: 第一步：检查遥控器是否有报警，按照对应故障码处理；第二步：检查手刹、转向是否正常，确认低压供电正常；第三步：检查12V蓄电池是否亏电；第四步：检查保险盒，确认是否有保险丝烧断；第五步：检查制动继电器，确认是否烧坏，可交叉验证。

Step 1: Check whether the remote controller has an alarm, and deal with it according to the corresponding fault code. Step 2: Check whether the handbrake and steering are normal, and confirm that the low-voltage power supply is normal. Step 3: Check whether the 12V battery is under charge. Step 4: Check the fuse box to see if any fuse is blown. Step 5: Check the brake relay to confirm whether it is burnt out, which can be cross-validated.

Q: 底盘手刹失效 Chassis handbrake failure

A: 第一步: 检查遥控器是否有报警, 按照对应故障码处理; 第二步: 检查制动、转向是否正常, 确认低压供电正常; 第三步: 检查12V蓄电池是否亏电; 第四步: 检查保险盒, 确认是否有保险丝烧断; 第五步: 检查手刹继电器, 确认是否烧坏, 可交叉验证。

Step 1: Check whether the remote control has an alarm, and deal with it according to the corresponding fault code. Step 2: Check whether the braking and steering are normal, and confirm that the low-voltage power supply is normal. Step 3: Check whether the 12V battery is under charge. Step 4: Check the fuse box to see if any fuse is blown. Step 5: Check the handbrake relay to confirm whether it is burnt out, which can be cross-validated.

Q: CAN口通信异常 CAN port communication is abnormal

A: 第一步: 检查遥控器是否有报警, 确认底盘其他各项功能正常, VCU正常工作; 第二步: 用万用表电压挡分别测量CAN口的2和7口, 负极接底盘外壳, 正常情况下, 2口电压低, 约2.25V, 7口电压高, 约2.75V, 若电压正常, 请检查自行接的另一端CAN口接头; 第三步: 若电压异常, 沿着CAN线检查线路, 确认是否有短路或断路。

Step 1: Check whether the remote control has an alarm, confirm that other functions of the chassis are normal, and the VCU is working normally. Step 2: Use a multimeter to measure the CAN port 2 and 7 respectively, and connect the negative pole to the chassis shell. Under normal circumstances, The voltage of port 2 is low, about 2.25V, and the voltage of port 7 is high, about 2.75V. If the voltage is normal, please check the CAN port connector at the other end connected by yourself. Step 3: If the voltage is abnormal, check the line along the CAN line to confirm whether there is a short circuit or an open circuit.

Q: 遥控器启动异常处理 Remote control startup exception handling

A: 整车上电成功后遥控车显示屏中下端会显示Ready, 反之上电失败, 车辆进入Ready后, 观察Ready上方是否有三角感叹号。如果无, 驾驶者可以进行驾驶操作。如果出现说明车辆存在故障, 需要按Display Switch按钮, 把显示切换当第二界面, 查看具体系统故障。进一步排查详细故障需要把通过OBD接口连接到相系统的CAN网络中读取详细的故障代码, 排除故障后车辆才能整车驾驶。

After the vehicle is powered on successfully, the lower part of the display screen of the remote control car will display "Ready". Otherwise, the power on fails. After the chassis enters "Ready", observe whether there is a triangle exclamation mark above "Ready". If not, the driver can perform a driving operation. If it shows that there is a fault in the vehicle, you need to press the Display Switch button to switch the display as the second interface to check the specific system fault. Further investigation of detailed faults requires reading detailed fault codes from the CAN network connected to the phase system through the OBD interface, and the vehicle can be driven as a whole after troubleshooting.

以下为遥控器故障码分析及处理建议

Remote Controller Fault Code Analysis and Suggestions for Handling is below

序号 No	故障码 Error code	故障码中文 Fault code in Chinese	故障码含义 Meaning of failure code	处理建议 Handling suggestion
1	Chassis normal	系统正常 System is normal	正常状态 Normal status	正常状态 Normal status
2	Unable to run	无法运行 Can not operate	无法运行 Can not operate	1、检查动力电池剩余电量，若低于 20%， 请充电后再使用 Check the remaining power of the power battery, if it is less than 20%, please charge it before use 2、检查高压、低压保险盒，确认是否有保 险烧断 Check the high-voltage and low- voltage fuse boxes to confirm whether any fuse is blown; 3、检查手刹或刹车是否归位 Check that the handbrake or brakes are in place. 4、采集报文分析。Collect and analyze the messages
3	F-steer failure	前转向机故障 Front steering system fault	转向系统电机或 控制器故障 Steering system motor or controller fault	1、检查保险盒，确认是否有保险丝烧断 Check the fuse box to see if there is a blown fuse 2、检查转向电机继电器 Check steering motor relay 3、采集报文分析 Collect and analyze the messages
4	B-steer failure	后转向机故障 Back steering system fault		
5	FL-Motor failure	前左轮故障 Front Left wheel failure	电机驱动器无高 压输入、或信号 线通讯失败 The motor driver has no high- voltage input, or the communication of the signal line fails.	1、检查动力电池剩余电量，若低于 20%， 请充电后再使用 Check the remaining power of the power battery, if it is less than 20%, please charge it before use 2、检查高压、低压保险，电机唤醒继电器 Check the high-voltage and low-voltage fuses, and the motor wake-up relay 3、检查对应车轮电机控制器，高压输入是 否正常，信号输入是否正常。Check the corresponding wheel motor controller, whether the high voltage input is normal, and whether the signal input is normal. 4、采集报文分析 Collect and analyze the messages
6	RL-Motor failure	后左轮故障 Rear Left wheel failure		
7	FR-Motor failure	前右轮故障 Front Right wheel failure		
8	RR-Motor failure	后右轮故障 Rear Right wheel failure		

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9	Brake failure	制动系统故障 Brake system failure	制动系统刹车泵或控制器故障 Brake system brake pump or controller failure	1、遥控操作刹车，检查是否有刹车声音 Operate the brakes by remote control, check whether there is a brake sound 2、检查刹车油壶油量是否充足 Check whether the brake oil pot is sufficient 3、检查刹车各个卡钳位置油管是否漏油 Check whether the oil pipes at each caliper position of the brake are leaking oil 4、松开手刹，挂空挡，用手推车，确认底盘是否能动 Release the handbrake, put in neutral, and use a trolley to confirm whether the chassis can move 5、采集报文分析 Collect and analyze the messages
10	ModeSwitch failure	模式切换失败 Mode switch failure	底盘开动过程中切换转向模式或换挡	1、停下来切换转向模式，停下来换挡 Switching steering modes or shifting gears while the chassis is moving 2、采集报文分析 Collect and analyze the messages
11	GearSwitch failure	挡位切换失败 Gear switch failure	Switching steering modes or shifting gears while the chassis is moving	
12	Lost connection	无线通讯异常 Abnormal wireless communication	遥控器与接收器通讯异常 The communication between the remote control and the receiver is abnormal.	1、底盘未开机，或接收器未上电，底盘开机，或检查接收机供电是否正常。The chassis is not powered on, or the receiver is not receiving power. Turn on the chassis, or check whether the power supply of the receiver is normal 2、遥控器与接收器距离过长大于 100mm，检查接收机天线是否正常，或近距离操控。The distance between the remote control and the receiver is too long and greater than 100mm. Check whether the antenna of the receiver is normal, or operate it at close range. 3、遥控器与接收器之间有信号屏蔽或干扰，更换地点测试。There is signal shielding or interference between the remote control and the receiver, change the location to test.
13	Low battery	遥控器电池电量低 The battery of Remote control is low	遥控器电池电量低 The battery of Remote control is low	更换遥控器干电池 Replacing the dry battery of the remote control

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14	Loss of VCU signal	接收器 CAN 信号丢失 Receiver CAN signal lost	接收器 CAN 信号丢失 Receiver CAN signal lost	检查接收器 CAN 网络 Check the CAN network of receiver
15	Push rod back to center	推杆故障 摇杆回中位再启动 Push rod failure Return the push rod to the neutral position and restart	推杆损坏 摇杆未回中，自检不过 The push rod is not in the neutral position, and the self-test fails	确保推杆回中，重启遥控器，遥控器自检过程禁止推动推杆 Make sure the push rod is back to the neutral position(center), restart the remote control, and the push rod is prohibited during the self-test process of the remote control 若多次尝试还显示故障，联系技术处理 If the failure is still displayed after repeated attempts, please contact Pix Moving for support
16	ESTOP!	急停按钮激活 Emergency stop button activated	急停按钮被按下 Emergency stop button is pressed	急停按钮激活，要想控制车辆，请退出急停状态 The emergency stop button is activated. To control the chassis, please exit the emergency stop state
17	E1	无线通讯异常 Abnormal wireless communication	遥控器与接收器通讯异常。 The communication between the remote control and the receiver is abnormal.	1、底盘未开机，或接收器未上电，底盘开机，或检查接收机供电是否正常。The chassis is not powered on, or the receiver is not receiving power. Turn on the chassis, or check whether the power supply of the receiver is normal 2、遥控器与接收器距离过长大于 100mm，检查接收机天线是否正常，或近距离操控。The distance between the remote control and the receiver is too long and greater than 100mm. Check whether the antenna of the receiver is normal, or operate it at close range. 3、遥控器与接收器之间有信号屏蔽或干扰，更换地点测试。There is signal shielding or interference between the remote control and the receiver, change the location to test.
18	E2	遥控器电池电量低 The battery of Remote control is low	遥控器电池电量低 The battery of Remote control is low	更换遥控器干电池 Replacing the dry battery of the remote control

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19	E3	接收机 CAN 掉线 Receiver CAN disconnected	接收机与 VCU 之间通讯异常。 The communication between the receiver and VCU is abnormal.	1、底盘 12V 没电，VCU 没启动，测量底盘 12V 电池电压。The 12V battery of the chassis is out of power and the VCU is not activated. Measure the voltage of the 12V battery of the chassis. 2、底盘动力电池没电，VCU 没工作，给底盘动力电池充电。The chassis power battery is out of power , the VCU is not working.Please charge the chassis power battery. 3、接收机 CAN 线接触不良或线路异常，检查接收机排线，重新插拔测试，检查接收机到 VCU 的线路。The CAN line of the receiver is not in good contact or the line is abnormal. Check the receiver cable, re-plug and test, and check the line from the receiver to the VCU.
20	E4	遥控器推杆故障 Remote control push rod failure	遥控器推杆故障，自检未通过。 Remote control push rod failure. The self-test fails	1、遥控器开机自检过程中，推杆被推动，导致自检未通过，重启遥控器 During the power-on self-test of the remote control, the push rod is pushed, resulting in the failure of the self-test. Restart the remote control. 2、遥控器推杆本身故障，联系厂家技术处理。The push rod of the remote control itself is faulty, please contact the PIX Moving for technical solution.
21	E5	无线通信异常 Abnormal wireless communication		
22	E6	备用 Backup		

VII. 日常维护与注意事项Maintenance And

Precautions

7.1 底盘使用提示Chassis Use Tips

车辆运行时应该注意保护周围人员安全同时保护底盘避免受到伤害，运行底盘时应该保证底盘周围环境安全。应该遵循以下安全运行指导：

When the chassis is running, you should pay attention to protecting the safety of the surrounding people and protecting the chassis from injury. When running the chassis, you should ensure the safety of the environment around the chassis. The following guidelines for safe operation should be followed:



◆ 底盘周围禁止人员围观；

People around the chassis are prohibited from watching

◆ 底盘运行道路通畅；

The roads for chassis running should be kept smooth;

◆ 避免雨天使用底盘；

Avoid using the chassis in rainy days

◆ 避免-10℃环境下使用底盘；

Avoid using the chassis below -10℃

◆ 禁止小孩在底盘周围逗留玩耍；

Do not allow children to stay and play around the chassis

◆ 禁止在坡度大于20°的地方使用底盘；

It is forbidden to use the chassis at a place with a slope greater than 20°

◆ 禁止搭载超过底盘最大负重的货物；

It is forbidden to carry goods that exceed the maximum load of the chassis;

◆ 避免将底盘停放于坡度较大的位置；

Avoid parking the chassis at places with steep slopes;

◆ 避免人员站在底盘轴线；

Avoid people standing on the chassis axis;

◆ 在非紧急情况下，请勿将急停开关按钮长期处于按下状态。

In non-emergency situations, do not keep the emergency stop switch button pressed for a long time.

7.2 电池使用注意Battery And Charger Use Considerations

警告！(Warning!)

电池使用时警告及注意事项(Notice and warning items using the battery)

为防止电池可能发生泄漏,发热、爆炸,请注意以下预防措施: In order to prevent the battery leaking, getting hot and exploding, please pay attention to preventing measure as following:

◆ 严禁将电池浸入海水或水中,保存不用时,应放置于阴凉干燥的环境中;

It is strictly forbidden to immerse the battery in seawater or water. When not in use, it should be placed in a cool and dry environment.

◆ 禁止将电池在热高温源旁,如火、加热器等使用和留置;

It is forbidden to use and leave the battery next to heat and high temperature sources, such as fire, heaters, etc.

◆ 充电时请选用锂离子电池专用充电器;

When charging, please use a specific charger for lithium batteries.

◆ 严禁颠倒正负极使用电池;

It is strictly forbidden to reverse the positive and negative poles of the battery

◆ 严禁将电池直接插入电源插座;

Do not insert the battery directly into the power socket

◆ 禁止将电池丢于火或加热器中;

Do not throw batteries into fire or heater

◆ 禁止用金属直接连接电池正负极导致短路;

It is forbidden to directly connect the positive and negative poles of the battery with metal to cause a short circuit

◆ 禁止将电池与金属,如发夹、项链等一起运输或贮存;

It is forbidden to transport or store batteries together with metals, such as hairpins, necklaces, etc.

◆ 禁止敲击或抛掷、踩踏电池等;

It is forbidden to knock or throw, step on the battery, etc.

◆ 禁止直接焊接电池和用钉子或其它利器刺穿电池;

It is forbidden to directly weld the battery and pierce the battery with nails or other sharp objects;

◆ 禁止在强静电和强磁场的地方使用,否则易破坏电池安全保护装置,带来不安全的隐患

It is forbidden to use it in places with strong static electricity and strong magnetic field, otherwise it will easily damage the battery safety protection device and bring unsafe hidden dangers

◆ 若电池发生泄露,电解液进入眼睛,请不要揉擦,应用清水冲洗眼睛,并立即送医治疗,否则会伤害眼睛。

If the battery leaks and the electrolyte enters the eyes, please do not rub them. Rinse the eyes with clean water and send to a doctor immediately, otherwise it will hurt the eyes.

- ◆ 如果电池发出异味、发热、变色、变形或使用、贮存、充电过程中出现任何异常,立即将电池 从装置或充电器中移离并停用。

If the battery emits peculiar smell, heat, discoloration, deformation or any abnormality occurs during use, storage or charging, immediately remove the battery from the device or charger and stop using it.

- ◆ 如果电极弄脏,使用前应用干布抹净,否则可能会导致接触不良功能失效。

If the electrode is dirty, wipe it off with a dry cloth before use, otherwise it may cause poor contact and function failure.

- ◆ 废弃之电池应用绝缘纸包住电极,以防起火、爆炸。

Discarded batteries should be wrapped with insulating paper to prevent fire and explosion.

充电器 Charger:

- ◆ 请勿将充电机放置在雨淋位置!

Do not place the charger in the rain!

- ◆ 本机内有高压部件, 请勿擅自拆解充电机!

There are high-voltage components inside the charger, please do not disassemble it without authorization!

- ◆ 请勿阻塞充电机进风和出风口!

Do not block the air inlet and outlet of the charger!

- ◆ 蓄电池电压与充电机标称电压必须相符!

The battery voltage must match the nominal voltage of the charger!

- ◆ 移动充电机时, 请断开电源线和充电插头。

Disconnect the power cord and the charging plug while moving the charger.

- ◆ 如有疑问, 可致电公司, 我公司将竭诚为您服务。

If you have any questions, you can call the company, and our company will serve you wholeheartedly.

7.3 停放维护Parking Maintenance

长时间不使用底盘时, 请按照以下建议操作:

When the chassis will not be used for an extended period of time, follow these suggestions:

1. **车辆停稳拉手刹。** 车辆停放场地需为平坦场地 (禁止将车辆长时间停放在坡道上, 如因车辆故障等原因必须将车辆停放在坡道上时, 必须用三角挡块或者石块等坚硬物体挡在车轮下坡方向防止溜车), 车辆停稳后应拉起驻车制动手柄。

1. Stop the vehicle and apply the handbrake. The chassis parking site must be flat (it is forbidden to park the chassis on the slope for a long time. If the chassis must be parked on the slope due to chassis failure or other reasons, it must be blocked by triangular blocks or stones and other hard objects in the downhill direction of the wheels. To prevent the vehicle from rolling), the parking brake handle should be pulled up after the chassis stops.

2.做好防雨防潮防尘。 底盘电器件不能接触雨水、灰尘，车辆停放时要做好防雨，接口处做好防尘和绝缘。

2. Do a good job of rainproof, moistureproof and dustproof. The electrical components of the chassis should not be in contact with rainwater or dust. When the vehicle is parked, it should be protected from rain, and the interface should be protected from dust and insulation.

3.12V蓄电池下电。 长时间停放车辆时请将12V小电池下电-将电源负级从电池上卸下并做绝缘处理。

3. The 12V battery should be powered off. When the chassis is parked for a long time, please power off the 12V small battery. Remove the negative pole of the power supply from the battery and perform insulation treatment.

4.防锈。 用湿毛巾、毛刷等工具将车辆擦拭干净，必要的地方涂防锈油，进行日常例行保养。

4. Anti-rust. Wipe the vehicle clean with a wet towel, brush and other tools, and apply anti-rust oil where necessary for daily routine maintenance.

5.远离火源和低温。 当气温低于 0 °C度时，应将车辆停入室内，以免低温导致底盘无法启动。

5. Keep away from fire and low temperature. When the temperature is lower than 0 °C, the chassis should be parked indoors, so as not to cause the chassis to fail to start due to low temperature.

长时间不使用底盘需要启动底盘时，使用前请做一下操作：

When you need to start the chassis without using it for a long time, please do the following operations before use:

1. 检查电池电量是否满足使用要求。
Check whether the battery power meets the usage requirements.
2. 检查各油管接头有无漏油情况。
Check each oil pipe joint for oil leakage.
3. 检查蓄电池电极柱导线是否拧紧，其它导线有无松脱。
Check whether the wires of the battery electrode column are tightened and whether other wires are loose.
4. 检查轮胎气压（2.6-2.8bar）、转向、制动功能是否正常。
Check whether the tire pressure (2.6-2.8bar), steering and braking functions are normal.
5. 检查灯光等信号装置是否正常。
Check whether the lighting and other signal devices are normal.
6. 当发现悬架、制动、转向或连接处卡涩、锈蚀时应使用适量润滑油润滑。
When it is found that the suspension, brake, steering or connection are jammed or corroded, an appropriate amount of lubricating oil should be used for lubrication.
7. 检查电池电量，当车辆电池电量过低时应及时充电。
Check the battery level, and charge it in time when the vehicle battery level is too low.
8. 充电环境应保持干燥通风，车辆及充电设备周围2米范围内不能有可燃物。

The charging environment should be kept dry and ventilated, and there should be no combustibles within 2 meters around the vehicle and charging equipment.

9. 检查中所发现的问题，应在车辆运行前进行排除。

Problems found during the inspection should be eliminated before the vehicle runs.

7.4 定期保养建议Maintenance Advice

对底盘定期进行保养，有助于减少底盘零部件故障，延长底盘使用寿命。

Regular maintenance of the chassis will help reduce the failure of chassis components and prolong the service life of the chassis.

➤ 结构外观 Structural Appearance

关机清理底盘上的灰尘、污渍，防止进入内部腐蚀元器件；检查底盘车体结构，漆面正常无脱落、螺栓锁紧无松动、线束正常无断裂；给转向球头、悬架摆臂衬套、摆臂固定座等活动部位涂抹润滑脂；检查底盘引出给上装的线束，确保无破损、断裂等故障。

Turn off the chassis. Clean the dust and stains on the chassis, and prevent the components from entering the interior. Check the chassis body structure, the paint surface is normal without falling off, the bolts are locked without looseness, and the wiring harness is normal without breakage. Apply grease to the moving parts such as the steering ball joint, the suspension swing arm bushing, and the swing arm fixing seat. Check the wiring harness leading from the chassis to the bodywork to ensure that there are no faults such as damage or breakage.

➤ 转向系统 Steering System

检查转向拉杆螺母锁紧无松动、转向节臂紧固牢靠无松动；测试转向能正常转到最大角度，松开摇杆后自动顺畅返回中位；测试转向过程无异响，响应灵敏、顺畅。

Check that the steering tie rod nut is locked tightly without looseness, and the steering knuckle arm is fastened firmly without looseness. The test steering can be turned to the maximum angle normally, and it will automatically and smoothly return to the neutral position after releasing the joystick. There is no abnormal sound during the test steering process, and the response is sensitive and smooth.

➤ 刹车系统 Braking System

检查刹车卡钳无漏油，油管无破损、断裂；检查确认刹车油壶油量充足，建议一至两年更换一次刹车油；检查确认刹车盘无变形、破裂，刹车片消耗未到安全线；测试刹车功能正常，刹车无异响，松开刹车后驱动无摩擦、阻力。

Check that the brake calipers have no oil leakage, and the oil pipes are not damaged or broken. Check to make sure that there is enough oil in the brake oil pot. It is recommended to replace the brake oil once every one to two years. Check to confirm that the brake disc is not deformed or cracked, and the brake pads are not consumed to the safety line. Test that the brake function is normal, there is no abnormal sound when the brake is released, and there is no friction and resistance when the brake is released.

➤ 驱动系统 Power System

检查轮毂和轮胎外观无破损、变形，轮毂电机安装螺母紧固无松动；检查轮胎气压充足，轮胎花纹磨损未到安全线；测试电机转动顺畅无异响，每个轮子都正常转动。

Check that the appearance of the hub and tire is not damaged or deformed, and the mounting nuts of the in-wheel motor are fastened without loosening. Check that the tire pressure is sufficient, and the tire pattern wear has not reached the safety line. The motors under testing rotates smoothly without abnormal noise, and each wheel rotates normally.

➤ 手刹系统 Handbrake System

检查手刹卡钳安装紧固无松动，钢丝绳紧固无松脱；测试手刹松放正常，卡钳卡紧、回位正常；给卡钳活动弹簧部位涂抹润滑脂。

Check that the handbrake caliper is installed and fastened without looseness, and the wire rope is fastened without looseness. Test that the handbrake is released and released normally, and the caliper is clamped and returned normally. Apply grease to the movable spring part of the caliper.

➤ 遥控器 Remote Control

关机清理遥控器灰尘、污渍；检查遥控器电量，电量不足及时更换电池；检查遥控器摇杆、按键、拨杆，确保拨杆拨动、按键按压功能正常。

Turn off the chassis. Clean up the dust and stains on the remote control. Check the power of the remote control, and replace the battery in time if the power is low. Check the push rods, buttons, and levers of the remote control to ensure that the levers are moved and the buttons are pressed normally.

VIII. 售后与技术支持 After-sale Service & Technical Support

1. 本公司保证所提供货物质量合格。

PIX Moving guarantees that the quality of the goods provided is qualified.

2. 为保证您的使用，我们提供7×24小时服务：我们的客户服务工程师为您供全年365天、全天24小时的支持，快速响应您的服务请求，并与正确的部门一起为您提供支持。我们通过系统化的服务平台记录和跟踪您的服务请求，确保它们得到及时有效的处理。

To ensure your use, we provide 7×24 hours of service: our customer service engineers provide you with support 24 hours a day, 365 days a year, respond quickly to your service requests, and work with the correct department provide support for you. We record and track your service requests through a systematic service platform to ensure that they are dealt with in a timely and efficient manner.

3. 远程技术支持（质量问题，无论何时何地您的产品遇到突发或紧急情况，我们的技术支持工程师会通过可靠的网络连接，远程快速识别故障并提供解决方案，最大限度减少产品使用中断。

Remote technical support (quality problems, whenever and wherever your product encounters unexpected or emergency situations, our technical support engineers will quickly identify faults remotely and provide solutions through reliable network connections, minimizing product use interruption .

4. 设备除非由于重大质量问题导致无法使用，否则一经出售不得退货。对于已经验收的产品,在质保期内出现产品质量问题或损坏（非人为损坏）的，经本公司检验并确认确属产品质量问题的，本公司给予维修或更换，并承担相关费用。

The products cannot be returned once sold unless it is rendered unusable due to major quality issues. For products that have been accepted, if there is a product quality problem or damage (non-human damage) during the warranty period, if PIX Moving inspects and confirms that it is a product quality problem, PIX Moving will repair or replace it and bear the relevant costs.

5. 如果对产品是否存在质量问题产生争议的，可将争议产品交由中立、合法、专业的第三方检测机构进行检测（该检测机构必须先由双方共同确认）检测费用由提交方预交，最终由检测结果不利方承担。

If there is a dispute over whether the product has quality problems, the disputed product can be handed over to a neutral, legal, and professional third-party testing agency for testing (the testing agency must first be confirmed by both parties). The testing fee is prepaid by the submitter, and finally The party responsible for the unfavorable test results.

6. 产品超过质保期后造成的损坏或非因产品质量问题造成的损坏等，若需要维修，本公司有权进行评估并收取相应费用。

PIX Moving has the right to evaluate and charge corresponding fees if repairs are required for damages caused after the warranty period expires or damages not caused by product quality problems.

7. 若需要进行二次开发部分涉及软件开发、电气电路、机械结构等，需改变原有出厂配置原貌或对控制系统底层调试引起的非产品承诺技术指标，不归属为原产品质量问题。本公司可给予相关积极配合协助，但不承担事故责任。

it will not be attributed to the original product quality problems, if the secondary development (partly involving software development, electrical circuitry, mechanical structure, etc.) requires changes to the original factory configuration or to the control system underlying the debugging caused by technical indicators other than those promised by the product. PIX Moving can give relevant positive cooperation and assistance, but does not assume responsibility for the accident.

附加声明： Additional declaration.

- 1、本产品中自研软硬件产品的所有知识产权不随产品销售发生任何转让。

All intellectual property rights of the self-developed software and hardware products in this product are not transferred with the sale of the product.

- 2、未经翰凯斯授权，不得将有关产品的保密信息全部或部分透露给任何第三方。

Confidential information about the products must not be disclosed in whole or in part to any third party without authorization from PIX Moving.

- 3、若有上述违反知识产权保护的行为，翰凯斯将有权要求立即停止该侵权行为并依法追究其刑事及民事责任，追偿经济损失。

In the event of such a breach of intellectual property rights, PIX Moving will be entitled to demand that the infringement be stopped immediately and that criminal and civil liability and economic loss be pursued in accordance with the law.